Breathing: Because Chain homework

Year 8 Science · B4 Organisms · Breathing · Skill: Because Chain | About 30 minutes, by hand | Name: _____ Date: _____

The goal is the joining words: *so · which means · because.* Every box in a chain is caused by the one before it.

A · Build the chains [7 marks]

1. Breathing **in**. Fill the two empty boxes (2 marks). Hint: box 2 is about the *volume* of the chest; box 3 is about the *pressure*.

The diaphragm contracts and flattens, and the rib muscles lift the ribs	→		→		→	Air flows into the lungs

2. Breathing **out**. Fill in the three empty boxes (3 marks). Then write one **so** or **which means** sentence joining any two boxes (2 marks).

The diaphragm relaxes and domes up; the ribs move down and in	→		→		→	

B · The bell jar model [6 marks]

3. Say what each part of the model represents (4 marks).

Part of the model	What it represents in the body
Balloons	
Rubber sheet at the bottom	
Glass jar	
Y-shaped tube	

4. The rubber sheet is pulled down. Explain what happens to the balloons and why (2 marks).

C · Fix the answer [3 marks]

5. *Explain how air enters the lungs.* Zara wrote: **“The diaphragm moves down and the lungs get bigger. Air goes in because the lungs are bigger.”**

(a) Two links are missing. Name them (1 mark).

(b) Rewrite Zara's answer with the missing links, using *so* or *which means* (2 marks).

D · Full answer, by hand [6 marks]

6. Explain how the movement of the diaphragm and the rib muscles causes air to enter the lungs. *(Write it as you would in an exam.)*

Mark yourself	What it looks like
Level 1 (1–2 marks)	Steps listed or named, not joined (“the diaphragm moves, the ribs move, air goes in”).
Level 2 (3–4 marks)	Some steps joined, but a link is missing (often the pressure step).
Level 3 (5–6 marks)	All the steps in order and each one joined to the next: muscles → volume → pressure → air.

E · From last week [2 marks]

7. A boy sprints for a bus and starts to breathe much faster. Write a chain of at least **two** links explaining why.

Self-check (5 min)

- ☐ I used a joining word (**so** / **which means** / **because**) between every link.
- ☐ My chain for breathing in has **muscles → volume → pressure → air**, in that order.
- ☐ My last link is the result, not a repeat of the question.
- ☐ I marked myself against Level 1, 2 or 3 and said why.

Oral check (5 min, with a parent)

Say it out loud, from memory if you can. Parent asks: **“Walk me through breathing in, using ‘so’ or ‘which means’ each time.” (Without reading her answer.)**

Parent tick: ☐ answered without reading it ☐ answered with a prompt ☐ could not yet

Weekly review (for J, 5 min)

Which link did she miss most often?	
What was weak this week?	
What worked?	
Time taken (HTML / written / total)	
One change for next week	

Parent key and marking notes

1. Box 2: the volume of the chest cavity increases. Box 3: the air pressure inside the chest falls below the air pressure outside.

2. Box 2: the volume of the chest cavity decreases. Box 3: the pressure inside rises above the pressure outside. Box 4: air is pushed out of the lungs. The joining sentence needs *so* or *which means* between two boxes.

3. Balloons = lungs. Rubber sheet = diaphragm. Glass jar = chest cavity (ribcage). Y-tube = windpipe (trachea) and bronchi.

4. The volume inside the jar increases, so the pressure inside falls below the pressure outside, so air is pushed into the balloons and they inflate.

5. (a) Chest volume increases; pressure inside falls (below outside). (b) e.g. “The diaphragm moves down, so the volume of the chest increases. This makes the pressure inside fall below the pressure outside, so air is pushed in.” Zara's answer is circular: it says air goes in because the lungs are bigger, which only repeats the idea.

6. Look for: diaphragm contracts and flattens; rib muscles contract and lift the ribs; chest volume increases; pressure inside falls below the outside; air flows in. **Level 3** needs every step joined to the next. Missing the pressure step usually means Level 2. “The air is sucked in” on its own is not enough: it needs the pressure difference.

7. Muscles work harder, so cells respire faster and need more oxygen (and make more carbon dioxide), which means breathing rate and depth increase to deliver oxygen and remove carbon dioxide faster.

Why this matters: her science teacher said an 8 to a 9 depends on explaining answers better. That is this skill: not more facts, but the joining words. If she writes the right steps in the wrong order, that is useful information too: ask her to tap through the chain on screen again.